Enhancing Student Mental Health Support through Empathetic Social Robot Interactions

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Abstract

This project evaluates a social robot intervention designed to support university students' mental health during the transition to adulthood. Utilizing the Furhat robotic platform, we developed a "Weekly Mental State Check-in" application to act as a non-judgmental digital agent. The design was informed by semi-structured interviews with five students to identify common academic stressors and a preference for calm personas. We implemented two interaction conditions—Empathetic and Neutral—using the Furhat Python Realtime API. Experimental results from a between-subjects study ($N = 24$) demonstrate that multimodal emotional expressiveness significantly enhances perceived empathy and robot likability. Specifically, the empathetic condition showed a 35.4% increase in empathy ratings compared to the neutral baseline. The system also integrates a clinical-inspired safety protocol that monitors for high-risk keywords to trigger emergency clinical support.

SCHEMATIC SUMMARY

- **1) Objectives:** Bridge the student mental health treatment gap using an autonomous social robot; implement a sentiment-aware state-machine; and validate the impact of non-verbal empathy. These were achieved via autonomous dialogue management and a risk-detection safety mechanism.
- **2) Robot platform:** Furhat Social Robot (using the "Adult/Isabel" face texture).
- **3) UCD approach:** Semi-structured interviews with 5 university students (Leo, Faizan, Mia, Noah, Qida). Key findings: Strong preference for a calm, gender-neutral persona; identified 3 minutes as the ideal interaction length; academic deadlines and seasonal winter stress were the primary focus.
- **4) Emotional behaviours:** *Empathetic Condition:* Amazon Polly Joanna-Neural voice, dynamic micro-expressions, back-channel smile during listening, and sentiment-triggered gestures (Smile/BrowFrown). *Neutral Condition:* Salli voice, static face, and default idle behavior.
- **5) Experimental design:** Online between-subjects experiment; autonomous interaction via Python Realtime API; $N = 24$ participants (12 per condition).
- **6) Measures and analysis:** Likability via 5-point Godspeed Questionnaire [1]; self-developed 5-item PES adapted from Leite et al. (2013) and Paiva et al. (2017). Quantitative analysis performed using Independent-samples t-tests.
- **7) Main results:** Non-verbal cues significantly drive rapport; the empathetic condition scored 84.7% in empathy vs. 49.3% in neutral ($t(22) = 4.88, p = 0.0001$).
- **8) Use of generative AI:** ChatGPT was utilized for drafting localized interview scripts, debugging Python WebSocket connection issues, and refining technical report structure.

1 I. INTRODUCTION

- **Chu Wang**: Led the User-Centered Design (UCD) phase, responsible for conducting semi-structured interviews, analyzing user needs, and drafting the interaction scripts.
- **Yihang Feng**: Responsible for core system development, including the implementation of the dialogue state-machine logic using the Furhat SDK, and oversaw the Phase 3 statistical data processing and t-test analysis.

- **Ruiji Huang**: Led the empathy behavior design, developing and fine-tuning multimodal robotic behavior models, including visual assets (facial textures) and acoustic prosody.
- **Weixun Ma**: Managed the experiment execution phase, coordinating participant recruitment, online data collection workflows, and maintenance of experimental forms.

Additionally, the formulation of the project specification, preparation of the final presentation, and integration of the technical report were collaborative efforts completed by all team members.

1.1 B. Project Overview

Many university students face barriers to mental health support, including social stigma and long wait times for clinical services. Our project provides a scalable alternative through a Furhat robot that facilitates self-reflection. By focusing on the non-verbal channel, we investigate how robot prosody and facial movement can foster a safe space for students to disclose emotional states.

2 II. APPLICATION

2.1 2.1 A. Design of Application (UCD Synthesis)

Based on UCD interviews, we localized the application for Swedish student life. We selected the "Isabel" face texture to provide a young, non-threatening appearance. The script includes culturally resonant topics such as "fika," "Uppsala weather," and "dark winter days." We employed a non-probing, open-ended questioning style (e.g., "How would you describe...") to ensure the user felt respected. Participants explicitly noted that a robot feels "less judgmental" than a human when discussing academic failure.

2.2 2.2 B. Development of Application

The application is built on the Furhat Python Realtime API. The core logic is managed by two entry points: condition1_empathetic.py and condition2_neutral.py. A robust state-machine manages the flow: Greeting → Ask_Questions → Closing. The system implements real-time keyword spotting to analyze user sentiment. In the empathetic condition, negative sentiment (e.g., "stressed") triggers a BrowFrown and the expressive Joanna-Neural voice. In the neutral condition, the Salli voice delivers identical verbal content but with a static face. The **Safety Protocol** scans for risk keywords like "suicide," providing the Uppsala Student Health Service hotline upon detection.

3 III. METHODOLOGY

To isolate the independent variable (non-verbal cues), we conducted an online experiment where the verbal script remained 100% identical for both groups. $N = 24$ participants were randomly assigned to either condition. Participants viewed a first-person video of the interaction to ensure high visibility of the robot's micro-expressions and nodding patterns. We used the Godspeed Questionnaire to measure Likability [1] and the Perceived Empathy Scale to assess emotional resonance [2].

4 IV. RESULTS

The results confirm that the empathetic non-verbal configuration significantly outperformed the neutral baseline across all metrics. Scores were normalized to percentages for readability.

- **Perceived Empathy:** The empathetic condition achieved a score of 84.7%, while the neutral baseline scored 49.3%. This **35.4% absolute increase** was statistically significant ($t(22) = 4.88, p = 0.0001$).
- **Likability:** The empathetic robot scored 82.7% in likability compared to 45.0% in the neutral condition ($t(22) = 5.26, p = 0.0000$). This represents a **37.6% improvement** in user acceptance.
- **Safety Trigger:** The safety protocol successfully detected predefined risk keywords during internal testing, validating the system's safety-conscious design.

$$\text{Percentage Score} = \left(\frac{\text{Mean Score}}{\text{Max Possible Score}} \right) \times 100\% \tag{1}$$

Table 1: Experimental Results

Metric	Empathetic	Neutral	Statistical Test
Perceived Empathy	84.7%	49.3%	$t(22) = 4.88, p = 0.0001$
Likability	82.7%	45.0%	$t(22) = 5.26, p = 0.0000$

Both scales exhibited excellent internal consistency (Godspeed Likability $\alpha = 0.96$, PES $\alpha = 0.97$). The effect sizes were very large for both Perceived Empathy (Cohen's d = 1.99) and Likability (d = 2.15).

5 V. DISCUSSION

5.1 A. Interpretation of Results

The 35.4% gap in empathy scores suggest that for students, the robot's "Expressive Layer" is more influential than the words spoken [3]. The Joanna-Neural prosody provided a human-like cadence that made users feel "heard," whereas the Salli voice was perceived as dismissive despite saying identical supportive phrases.

5.2 B. Reflection on Failures and Limitations

Several areas for improvement were identified during the study:

- **Sentiment Analysis Accuracy:** Our keyword-based NLP model struggled with sarcasm. Future iterations should use LLM-based sentiment analysis for higher accuracy.
- **Latency Issues:** The WebSocket connection occasionally introduced a 1-second delay in gestures, occasionally disrupting the "empathy-loop."
- **Immersion:** Video-based testing lacks physical presence. Participants mentioned they would feel more accountable if the robot were physically in the room.

5.3 C. Ethical issues / Societal Impact

The project reduces the stigma of seeking help by providing an anonymous digital first point of contact. However, we must ensure that the robot is never viewed as a clinical replacement for a human therapist. Its role is strictly for screening and self-reflection.

6 VI. GENERATIVE AI USE

Generative AI (ChatGPT) was instrumental in this project. It was used for iterative prototyping support, debugging WebSocket integration, and improving the clarity of technical writing. This allowed the team to focus on the HRI behavioral design and behavioral assets.

7 VII. CONCLUSION

This project demonstrates that non-verbal expressivity is a functional necessity for mental health social robots. The significant 35.4% increase in perceived empathy validates that multimodal cues are the primary drivers of trust. By integrating safety protocols and localization, we have created a viable prototype for autonomous student support.

References

[1] Bartneck, C., Kulić, D., Croft, E., & Zoghbi, S. (2009). Measurement instruments for the anthropomorphism, animacy, likeability, perceived intelligence, and perceived safety of robots. International Journal of Social Robotics, 1(1), 71-81.

[2] Leite, I., Castellano, G., Pereira, A., Martinho, C., & Paiva, A. (2013). Empathic agents for long-term interaction. International Journal of Social Robotics, 5(3), 329-341.

[3] Paiva, A., Leite, I., Boukricha, H., & Wachsmuth, I. (2017). Empathy in virtual agents and robots: A survey. ACM Transactions on Interactive Intelligent Systems (TiIS), 7(3), 1-40.

APPENDIX

The complete source code for the Furhat robot application, the raw experimental dataset, and the Python scripts used for statistical analysis are all available in our project's GitHub repository: https://github.com/abandon1232/HRI-Project-Group21.git.

A Project Specification and UCD Summary

A.1 Project Specification Summary

Goal: To design and evaluate a Social Robot intervention for student mental health check-ins.
Technical Features: State-machine dialogue flow, real-time sentiment analysis, and a clinical-inspired safety protocol (detecting suicidal ideation).
Target User: Uppsala University students (18-25 years old) facing academic stress.

A.2 UCD Interview Synthesis

We interviewed five students (Leo, Faizan, Mia, Noah, Qida).

- **Stressors:** Exam performance, group projects, and winter loneliness.
- **Persona preference:** Gentle, calm, and gender-neutral faces were preferred to avoid the uncanny valley.
- **Duration:** 3 minutes was identified as the optimal length for a recurring weekly session.

B Experimental Questionnaire

B.1 Godspeed Questionnaire (Likability)

Please rate the robot on a scale of 1 to 5 for the following pairs: 1. Unfriendly / Friendly; 2. Unkind / Kind; 3. Unpleasant / Pleasant; 4. Awful / Nice; 5. Dislike / Like.

B.2 Perceived Empathy Scale (PES)

Rate your agreement (1 = Strongly Disagree, 5 = Strongly Agree): 1. The robot seemed to understand how I was feeling. 2. The robot showed empathy toward me. 3. The robot's reactions were appropriate for my emotions. 4. I felt the robot cared about my well-being. 5. The robot's non-verbal feedback (e.g., nodding) made me feel listened to.

C original project's specifications/proposal

C.1 INTRODUCTION

This project aims to design and evaluate a Social Robot intervention to support university students' mental health during the challenging transition into adulthood[cite: 128]. Using the Furhat robotic platform, we will develop a "Weekly Mental State Check-in" application[cite: 129]. The robot acts as an empathetic interviewer, conducting 3-minute sessions focused on students' feelings, activities, and social connections[cite: 130]. The project addresses the "treatment gap" in student mental health by providing a low-barrier, non-judgmental digital agent for self-reflection and emotional tracking[cite: 131].

C.2 OBJECTIVES

- **UCD Integration:** Apply User-Centered Design methods to create an interview script that resonates with the typical student experience[cite: 136].
- **System Development:** Implement two distinct interaction conditions (Empathetic vs. Neutral) using the Furhat SDK, focusing on facial textures and speech prosody[cite: 137].

- **Experimental Validation:** Conduct a controlled HRI experiment with $N \geq 20$ participants to test whether emotional expressiveness significantly enhances perceived empathy[cite: 138].
- **Ethical Safety Mechanism:** Design and implement a strict "Safety Protocol" closed-loop system inspired by clinical AI guidelines[cite: 140].

C.3 METHODS

- **User-Centered Design (UCD):** Conduct semi-structured interviews with 5 university students to identify common stressors and preferred robotic personas[cite: 144].
- **Technical Implementation:** Condition 1 (Empathetic) utilizes dynamic face textures, increased nodding, and modulated speech prosody[cite: 149]. Condition 2 (Neutral) employs default idle behavior and monotonic voice[cite: 150].
- **Experimental Design:** An online between-subjects experiment where participants view a first-person video and complete a survey incorporating the Godspeed Questionnaire and PES[cite: 151, 152].

C original project's specifications/proposal

I. INTRODUCTION

This project aims to design and evaluate a Social Robot intervention to support university students' mental health during the challenging transition into adulthood. Using the Furhat robotic platform, we will develop a "Weekly Mental State Check-in" application. The robot acts as an empathetic interviewer, conducting 3-minute sessions focused on students' feelings, activities, and social connections. The project addresses the "treatment gap" in student mental health by providing a low-barrier, non-judgmental digital agent for self-reflection and emotional tracking. To address these needs, we designed a Furhat social robot with a calm, gender-neutral, young face texture and gentle speech prosody to reduce stigma and avoid the uncanny valley.

II. OBJECTIVES

- **UCD Integration:** Apply User-Centered Design methods to create an interview script that resonates with the typical student experience.
- **System Development:** Implement two distinct interaction conditions (Empathetic vs. Neutral) using the Furhat SDK, focusing on facial textures and speech prosody.
- **Experimental Validation:** Conduct a controlled HRI experiment with $N \geq 20$ participants to test whether emotional expressiveness significantly enhances perceived empathy.
- **Academic Delivery:** Submit a technical report and deliver a presentation detailing the design, implementation, and statistical findings.
- **Ethical Safety Mechanism:** Design and implement a strict "Safety Protocol" closed-loop system. If high-risk keywords (e.g., suicidal ideation) are detected, the system will instantly halt and trigger an emergency intervention response providing the Student Health Service hotline.

III. METHODS

- **User-Centered Design (UCD):** Conduct semi-structured interviews with 5 students to identify stressors and preferred robotic personas, informing the "4 Ws" and dialogue script.
- **Technical Implementation:** Use Furhat SDK to build a state-machine dialogue system.
  - *Condition 1 (Empathetic):* High expressivity using dynamic face textures, increased nodding, and modulated speech prosody.
  - *Condition 2 (Neutral):* Baseline version with idle behavior, static expressions, and monotonic voice.

- **Experimental Design:** Online between-subjects experiment. Participants view a first-person video and complete the Godspeed Questionnaire and Perceived Empathy Scale.
- **Data Analysis:** Independent-samples t-tests to compare the two conditions.
- **Localization:** Script is culturally adapted to Swedish student life (e.g., including "fika").

IV. DELIVERABLES

- Project Specification (PDF): April 16th.
- Functional Furhat Application: finalized by Week 19.
- Final Presentation: May 20th.
- Final Project Report: May 22nd.

V. POTENTIAL ISSUES

- **SDK Learning Curve:** Mastering the State Machine (Mitigation: early prototyping).
- **Participant Engagement:** Ensuring survey responses (Mitigation: student social networks).
- **Ecological Validity:** Online interaction vs. in-person (Mitigation: high-quality recording).

VI. PROJECT BREAKDOWN

Deadline	Task Description	Assigned To
April 16	Submit Project Specification	All members
April 20	UCD Research: Interviews & Scripting	Chu Wang
April 27	Phase 2: Dialogue State Machine Coding	Yihang Feng
May 04	Phase 2: Empathy Behavior Design	Ruiji Huang
May 11	Phase 3: Experiment Execution (Online)	Weixun Ma
May 18	Phase 3: Data Analysis & T-testing	Yihang Feng
May 20	Final Presentation	All members
May 22	Final Report Submission	All members

VII. AIMED GRADE

Our team is aiming for Grade 5. We are committed to exceeding basic requirements by implementing a high-fidelity multimodal interaction model and conducting a rigorous scientific evaluation to ensure the highest quality of work.